Best RNN+CNN combination in image captioning

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Abstract

With the rapid development of deep learning, more and more scholars and researchers focus on image captioning, CNN+RNN is one of the greatest ways to solve image captioning. However, there is insufficient detailed research about which combination of CNNs and RNNs performs better for the CNN+RNN model. So the point of this article is to preliminary fill in the gaps in this section(Only the flicker8k dataset was considered). This article will mainly begin this study through a series of quantitative experiments and illustrate briefly the elaboration of the corresponding CNN pre-trained model and RNN structures. This essay will significantly compare, in terms of the model's loss function, the combination of mainstream pre-train CNN models (DenseNet201, ResNet50, VGG16, InceptionV3) and two types of RNN (LSTM, GRU) based on the filcker8K dataset. Finally, the conclusion of this essay is that the DenseNet structure is better than other structures for image capture and LSTM is worse than GRU in generating text.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Natural language processing; Information extraction.

Keywords

deep learning, image captioning, machine learning

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1 Introduction

In recent years, there has been a growing interest in image captioning, the definition of image captioning traditionally is that the model can generate a detailed text description based on the image [1]. Image captioning can help people generally understand the meaning of the picture and the blind and visually impaired to generally understand picture content. However, if the model wants to have great performance, the model not only needs the ability to understand the features of the picture like the human, recognizing the objects in the picture, such as animals, but also needs to understand what is happening in the picture. Oriol and Alexander mentioned that the NIC model(CNN+RNN) is a fantastic way to solve this problem [1]. According to their idea, the image is firstly extracted into a fixed-length vector representation by the CNN model and then fed into the decoder RNN to generate sentences. However, they did not focus on which combination works best with different CNN and RNN pre-trained models. There is still no idea about which the combination of CNN pre-trained model and RNN models is the best choice. So the goal of this paper is to find the best CNN pre-training model in terms of image captioning by comparing the loss function of different pre-training +RNN models through some series of experiments for quantitative and model comparison.

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2 Background

2.1 CNN

CNN(Convolutional Neural Network) is a kind of deep learning model, being used to deal with pictures, such as Image Recognition, Object Detection, and Image Segmentation [2]. This article focuses on designing four pre-trained CNN model architectures(DenseNet, VGG, MobileNet, and ReseNet). this four structures have their unique characteristics. Firstly, the DenseNet structure is tightly connected, which means DenseNet'each layer is directly connected to all preceding layers [3]. Secondly, the VGG structure consists of a stack of multiple convolutional blocks [4], which means it is easy to implement and understand. Thirdly, MobileNetV2 has an inverted residual block and a linear bottleneck design [5]. Fourthly, ReseNet introduces a new block, which is called a residual block [4]. Pre-trained models of these four architectures are cited in this paper for comparison. Although the experiments will must be affected by the number of parameters of the pre-trained model and the complexity of the model resulting in experimental results that do not directly reflect which of the pre-trained model's architectures is better, the experiments in this paper will try my best to eliminate these factors as experimental conditions.

2.2 RNN

RNN(Recurrent Neural Network) is a type of deep learning model,which is used to deal with sequence data, such as time series and text data [2]. LSTM has a cellular structure and gate mechanism, which makes LSTM powerful in processing sequence data [6]. GRU is a variant of LSTM, which has only two gates (reset gate, update gate) compared to three gates (input gate, forget gate, output gate) of LSTM [7]. CNN is used for image feature extraction, CNN will deal with the input image and extract the features from the image [1]. RNN can store and learn the temporal sequence of the input sequence. In image annotation, RNN is responsible for

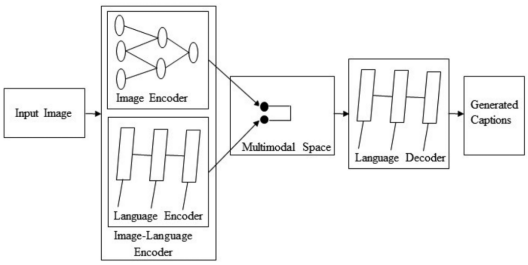

Figure 1: the process of multimodal space-based image captioning [11].

Table 1: An example of datasets.

Picture info	Picture name	Picture caption
	_6910002682013b08cb0e.jpg	1. A child in a pink dress is climbing up a set of stairs in an entry way . 2. A girl going into a wooden building . 3. A little girl climbing into a wooden playhouse . 4. A little girl climbing the stairs to her playhouse . 5. A little girl in a pink dress going into a wooden cabin .

generating the word sequence of the image based on the features of CNN [1].

2.3 Previous work

Image captioning has been implemented by many people before, especially for deep learning. Figure 1 shows the multimodal space-based image captioning [11].

3 Methods

3.1 Experimental principle

The idea of this essay is to dividing the image captioning model into two parts(CNN parts, RNN parts), trying to replace the CNN part and the RNN part separately. Depending on the same dataset(flicker8k), by replacing the CNN part with different pre-trained CNN models and the RNN part with LSTM and GRU, we compare their loss functions of the Image captioning models with these different CNN models. In the most situations, the loss function of the model is lower, then the model is better. And

3.1.1 Experimental Configuration. OS system: Microsoft Windows 11

CPU: Processor AMD Ryzen 9 5900HX with Radeon Graphics, 3301 Mhz, 8 cores, 16 logical processors

RAM: 16GB DDR4

GPU: NVIDIA GeForce RTX 3060 Laptop GPU

3.2 Dataset

3.2.1 Data collection. Flicker8k features 8,000 images from the Flicker website(the image content is usually about people and animals, while each image will have five captions describing the image content. In total, the image content has a total of 5,000 images and 40,000 sentences. The dataset consists of two parts, the images and the corresponding captions. Table 1 shows one item of the dataset.

3.2.2 Image part. This paper referenced this site's code for this part [8] and the architecture references this article [9].

This illustration shows an example of DenseNet201+LSTM, on which this paper will be modified.

Data preprocessing. All the images are converted to 224x224x3 pixel format.

Meaning. Determine all the images of the training set at 224x224 pixels and also specify the loading mode as RGB (colour pattern), this will ensure that all the images input to the neural network have the same shape and will be used as the same experimental condition.

3.2.3 Caption part. This paper referenced this site's code for this part [8] and the architecture references this article [9].

Data preprocessing. Convert all letters to lowercase, remove non-alphabetic symbols, replace multiple spaces with single spaces, remove single-length words, and add start and end characters. A vocabulary list is created, the longest sequence length is determined, the image captions are converted into a list, the length of this list is the length of the longest sentence, and then each word of the CAPTION is converted into an index in the corresponding vocabulary list. From there, all English sentences are converted into an array of integers

Meaning. Doing this transforms the text into patterns (numerical values) that the model can process, neural network models are essentially numerical processing, and after the transformation the neural network model will be able to process the sentences, and it

Table 2: The setting of experiments.

Data set partitioning	designated 85% of the data as the training set and 15% as the test set. Additionally, I planned to retain 85% of the data after each epoch to avoid overfitting. It is crucial to ensure consistent results, therefore I did not disturb the test set.
Loss function	categorical_crossentropy
Optimizer	Adam
Training batches	64
Number of training rounds	500
Learning rate	Adam optimizer defaults to 0.001, if the loss rate does not improve in three consecutive epochs, the learning rate will be reduced by 20%, and the lower limit of the learning rate will not be lower than 0.00000001.

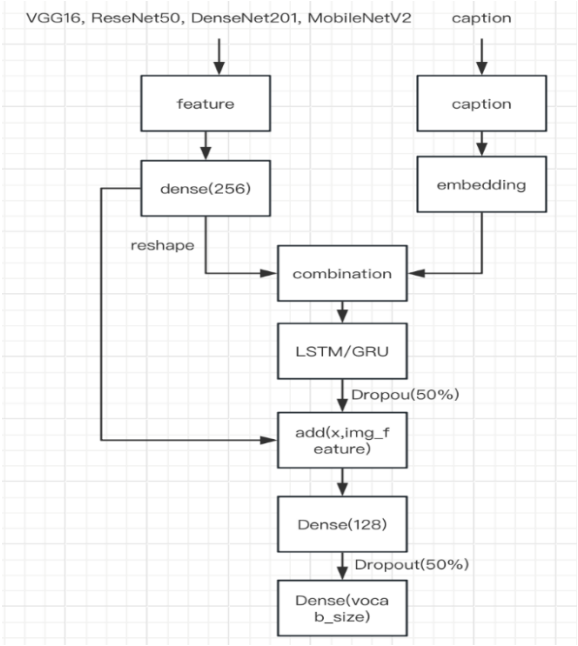

Figure 2: This table illustrates the structure of this model.

will also help the model to understand and process natural language. If this is not done it will be difficult to capture the relationship between the semantics and unable to train the model.

3.2.4 Experimental Essential information. Table 2 shows some impartial information about the initial setting.

3.3 Model architecture

This paper referenced this site's code for this part [8] and the architecture references this article [9]. Figure 2 illustrates the architecture of the project model.

How to experiment:

Comparing different combinations of RNNs and CNNs. Compare the results by depicting the difference between the number of training and the loss function.

4 Result

4.1 Experimental results

Table 3 illustrates the comparison of values between different models.

4.2 Result analysis

This essay mainly focuses on three norms(Learning rate, validation loss, and training loss). Table 4 explain three concepts which are important to analysing the results.

For learning rates(Select the first epoch to experience a drop in learning rate):

ResNet50+GRU(Flicker8k)>DenseNet121+GRU(Flicker8k)> DenseNet121+LSTM(Flicker8k)=VGG16+GRU(Flicker8k) =MobileNetV2+GRU(Flicker8k) =MobileNetV2+LSTM(Flicker8k) =ResNet50+LSTM(Flicker8k) =VGG16+LSTM(Flicker8k)

In terms of the speed of model convergence, ResNet50+GRU is the best combination, because Learning rates begin to decline at the earliest.

For validation loss(Select the smallest val-loss of all epochs):

DenseNet121+GRU(3.5741)>ResNet50+GRU(3.5836)>MobileNetV2 +GRU(3.5982)>ResNet50+LSTM(3.6156)>MobileNetV2+LSTM(3.6250) >DenseNet121+LSTM(3.6292)>

VGG16+GRU(3.6678)>VGG16+LSTM(3.7112)

In terms of Model performance on unseen data, DenseNet121+GRU is the best model.

For training loss(Select the last epoch loss):

DenseNet121+GRU(3.0805)>ResNet50+LSTM(3.0907) =ResNet50+LSTM(3.0907)> ResNet50+GRU(3.1047)> DenseNet121+LSTM(3.1841)>MobileNetV2+GRU(3.1829)> VGG16+GRU(3.2187)>VGG16+LSTM(3.2524)

In terms of Model performance on seen data, DenseNet121+GRU is the best model.

Overall, in this paper, validation loss is chosen as the only norm because validation loss means the generalization ability of the model, which usually is the most important ability.

5 Conclusion

In terms of the combination of CNN and RNN, the essay concludes that the best pre-trained CNN model for image capture is the DenseNet structure and GRU is better than LSTM in generating text.

Table 3: Analyses the results of the experiments.

Type	trend	Numerical changes
ResNet50+GRU(Flicker8k)	A quick decrease could be found in the loss function between 0 epoch and 6 epoch. After that, a slow decrease from epoch 6 to epoch 11.	Learning rate: 1-9: 0.001-0.001 10-11:2.0000e-04 validation loss: 1-6:4.0543 - 3.5836 6-11:3.5836 - 3.6196 training loss: 1-11:5.5510 - 3.1047
ResNet50+LSTM(Flicker8k)	A quick decrease could be found in the loss function between 0 epoch and 1 epoch, after that, a slow decrease from epoch 1 to epoch 12.	Learning rate: 1-11: 0.0010 12-13:2.0000e-04 validation loss: 1-8:4.1478 - 3.6029 9-13:3.6111 - 3.6453 training loss: 1-13:5.6476 - 3.0907
MobileNetV2+LSTM(Flicker8k)	A quick decrease could be found in the loss function between 0 epoch and 1 epoch, after that, a slow decrease from epoch 1 to epoch 12.	Learning rate: 1-11: 0.001-0.001 12-13:2.0000e-04 validation loss: 1-8:4.1755 - 3.6250 9-13:3.6468 - 3.6929 training loss: 1-13:5.6455 - 3.0945
MobileNetV2+GRU(Flicker8k)	A quick decrease could be found in the loss function between 0 epoch and 1 epoch, after that, a slow decrease from epoch 1 to epoch 10.	Learning rate: 1-11: 0.001-0.001 11-12:2.0000e-04 validation loss: 1-7:4.0796 - 3.5982 8-11: 3.6100 - 3.6905 training loss: 1-12:5.6132 - 3.0440
VGG16+LSTM(Flicker8k)	A quick decrease could be found in the loss function between 0 epoch and 1 epoch, after that, a slow decrease from epoch 1 to epoch 17 epoch	Learning rate: 1-16: 0.001-0.001 17-18: 2.000e-04 validation loss: 1-8:4.4.3690 - 3.7313 9-18:3.7280 - 3.7506 training loss: 1-18:5.7917 - 3.2524
VGG16+GRU(Flicker8k)	A quick decrease could be found in the loss function between 0 epoch and 1 epoch, after that, a slow decrease from epoch 1 to epoch 12 epoch	Learning rate: 1-11: 0.0010 12-13:2.000e-04 validation loss: 1-8:4.2237-3.6595 9-13:3.6642-3.6791 training loss: 0-13:5.7249-3.2187
DenseNet121+LSTM(Flicker8k)	A quick decrease could be found in loss function between 0 epoch and 1 epoch, after that, a slow decrease from epoch 1 to epoch 12 epoch	Learning rate: 1-11: 0.001-0.001 12-13:2.0000e-04 validation loss: 1-8:4.2086-3.4463 9-13:3.6470-3.6443 training loss: 1-13:5.6562 - 3.1841
DenseNet121+GRU(Flicker8k)	A quick decrease could be found in loss function between 0 epoch and 1 epoch, after that, a slow decrease from epoch 1 to epoch 10 epoch	Learning rate: 1-10: 0.001-0.001 11-12:2.0000e-04 validation loss: 1-7:4.0650-3.5741 8-12:3.5897-3.6241 training loss: 1-12:5.5685-3.0805

Table 4: This table mainly talks about the meanings of three norms.

norm	meaning
Learning rate	The learning rate corresponds to the speed of model convergence
validation loss	Model performance on unseen data
training loss	Model performance on seen data

The meaning of this essay is to find the great combination of CNN and RNN on ficker8k. However, this paper fails to eradicate disturbances such as the number of parameters of the pre-trained model. This is consistent with the papers of other scholars, which increases the credibility of this paper [10]. In the future, more searches need to be done in this field and this article do not cover the attention mechanism.

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